In [1]:
```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score,precision_score,recall_score,f1_score
```

In [2]:
```
df=pd.read_csv('/home/backup/ml_datasets/wine_quality_red.csv')
print(df.head())
```

```
   fixed acidity  volatile acidity  citric acid  residual sugar  chlorides  \
0            7.4              0.70         0.00             1.9      0.076
1            7.8              0.88         0.00             2.6      0.098
2            7.8              0.76         0.04             2.3      0.092
3           11.2              0.28         0.56             1.9      0.075
4            7.4              0.70         0.00             1.9      0.076

   free sulfur dioxide  total sulfur dioxide  density    pH  sulphates  \
0                 11.0                  34.0   0.9978  3.51       0.56
1                 25.0                  67.0   0.9968  3.20       0.68
2                 15.0                  54.0   0.9970  3.26       0.65
3                 17.0                  60.0   0.9980  3.16       0.58
4                 11.0                  34.0   0.9978  3.51       0.56

   alcohol  quality
0      9.4        5
1      9.8        5
2      9.8        5
3      9.8        6
4      9.4        5
```

In [3]:
```
X=df.drop('quality',axis=1)
y=df['quality']
```

In [4]:
```
y=(y>=6).astype(int)
print(" class distribution:\n")
print(y.value_counts())
```

```
 class distribution:

1    855
0    744
Name: quality, dtype: int64
```

In [6]:
```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42,str
```

In [7]:
```
scalar=StandardScaler()
scalar.fit(X_train,y_train)
X_train=scalar.fit_transform(X_train)
X_test=scalar.transform(X_test)
```

In [14]:
```
models={
    "1 Hidden Layers-10 Neurons":(10,),
    "1 Hidden Layers-50 Neurons":(50,),
    "2 Hidden Layers-50,25 Neurons":(50,25),
    "2 Hidden Layers-100,50 Neurons":(100,50),
    "3 Hidden Layers-100,50,25 Neurons":(100,50,25),
```

```
    "3 Hidden Layers-100,100,50 Neurons":(100,100,50)
}
```

In [19]:

```
results=[]
for name,architecture in models.items():
    print("Training:",name)
    mlp=MLPClassifier(
        hidden_layer_sizes=architecture,
        activation='relu',
        solver='adam',
        max_iter=1000,
        random_state=42
    )
    mlp.fit(X_train,y_train)
    y_pred=mlp.predict(X_test)
    accuracy=accuracy_score(y_test,y_pred)
    precision=precision_score(y_test,y_pred)
    recall=recall_score(y_test,y_pred)
    f1=f1_score(y_test,y_pred)

    results.append([name,accuracy,precision,recall,f1])
```

```
Training: 1 Hidden Layers-10 Neurons
Training: 1 Hidden Layers-50 Neurons
/usr/local/lib64/python3.6/site-packages/sklearn/neural_network/_multilayer_perceptro
n.py:617: ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached
and the optimization hasn't converged yet.
  % self.max_iter, ConvergenceWarning)
Training: 2 Hidden Layers-50,25 Neurons
Training: 2 Hidden Layers-100,50 Neurons
Training: 3 Hidden Layers-100,50,25 Neurons
Training: 3 Hidden Layers-100,100,50 Neurons
```

In [23]:

```
results_df=pd.DataFrame(results,columns=['architecture','accuracy','precision','reca
print("Performance Comparison:\n",results_df.to_string(index=False))
```

```
Performance Comparison:
                       architecture  accuracy  precision    recall        f1
         1 Hidden Layers-10 Neurons  0.756250   0.807947  0.713450  0.757764
         1 Hidden Layers-50 Neurons  0.809375   0.857143  0.771930  0.812308
      2 Hidden Layers-50,25 Neurons  0.784375   0.800000  0.795322  0.797654
     2 Hidden Layers-100,50 Neurons  0.796875   0.819277  0.795322  0.807122
  3 Hidden Layers-100,50,25 Neurons  0.790625   0.798851  0.812865  0.805797
 3 Hidden Layers-100,100,50 Neurons  0.790625   0.817073  0.783626  0.800000
```

In [24]:

```
best_model=results_df.loc[results_df['f1'].idxmax()]
print("Best Model:\n",best_model)
```

```
Best Model:
 architecture    1 Hidden Layers-50 Neurons
accuracy                          0.809375
precision                         0.857143
recall                             0.77193
f1                                0.812308
Name: 1, dtype: object
```

In [ ]: